

Name
Date
Course

Beginning Algebra Practice Test II (Version 2)

Instructions

- i) As always, write a title, name, date, and course.
- ii) Number each problem, circle the number, and show work when necessary.
- iii) Draw a rectangle around your answers.
- iv) Staple a copy of these test questions with your work and your final answers.

Simplify.

$$\textcircled{1} \frac{x^2 + 10x + 16}{x^2 - 64}$$

$$\textcircled{2} \frac{4x^2 - 9}{2x + 3}$$

$$\textcircled{3} \frac{10x^9 - 90x^7}{8x^7 - 40x^6 + 48x^5}$$

Perform the indicated operations.

$$\textcircled{4} \frac{x-2y}{x^2-25} \cdot \frac{x+5}{3x^2-14xy+16y^2} \quad \textcircled{5} \frac{x+4}{x^2-x} \div \frac{5x^2+20x}{x^2-2x+1}$$

$$\textcircled{6} \frac{3}{x+4} - \frac{7x}{x^2+8x+16}$$

$$\textcircled{7} \frac{2}{7x} + \frac{8x-5}{2x(x-6)}$$

Find the square roots. You will have to approximate one of the roots with a calculator (Round to the nearest thousandth).

⑧ a) $\sqrt{\frac{25}{49}}$

b) $\sqrt{64}$

c) $\sqrt{7}$

d) $\sqrt{9}$

⑨ Factor out a perfect square:

a) $\sqrt{60}$

b) $\sqrt{200}$

Add or subtract

c) $-8\sqrt{2} + 2\sqrt{2} + 3\sqrt{2}$

d) $3\sqrt{11} - \sqrt{11} - 8\sqrt{11}$

Multiply. Never leave perfect square factors under the root signs.

⑩ a) $\sqrt{5} \cdot \sqrt{5}$

b) $-5\sqrt{7} \cdot \sqrt{6}$

c) $2\sqrt{15}(-4\sqrt{2})$

d) $-3\sqrt{2}(5\sqrt{3} - 4\sqrt{2} + \sqrt{10})$

⑪ Divide or multiply.

a) $-2\sqrt{7}(-3\sqrt{5})$

b) $\frac{\sqrt{100}}{\sqrt{4}}$

Rationalize the denominators.

c) $\frac{3}{\sqrt{2}}$

d) $\frac{6}{\sqrt{40}}$

Simplify.

(12) a) $\frac{\frac{3}{4}}{\frac{8}{9}}$

b) $\frac{\frac{7}{5}}{8}$

c) $\frac{10}{\frac{3x}{8}}$

d) $\frac{\frac{2}{3} + \frac{1}{6}}{\frac{5}{9} - \frac{1}{2}}$

Divide.

$$(13) \frac{4x^4 - 8x^2 + 3}{16x^{12} + 24x^{10}} \div \frac{3 - 2x^2}{4x^4 - 9}$$

Simplify (i.e. add and/or subtract).

$$(14) a) \sqrt{32} - 8\sqrt{5} - 2\sqrt{3} - 6\sqrt{27}$$

Factor out a perfect square(s) (leave no perfect square factor in the root).

$$b) \sqrt{140}$$

Rationalize the denominators.

$$(15) a) \sqrt{\frac{1}{16}} \quad b) \sqrt{\frac{6}{7}} \quad c) \sqrt{\frac{8}{3}} \quad d) \sqrt{\frac{11}{16}}$$

Simplify.

$$(16) a) \frac{\frac{4}{x-3}}{\frac{2x}{3}}$$

Rationalize the denominator.

$$b) \sqrt{\frac{6}{63}}$$

Simplify.

$$(17) \frac{\frac{2x^2}{3y}}{\frac{5z}{8y^2}}$$

(18) Add and/or subtract.

$$\frac{3}{5y} - \frac{6}{y} + \frac{1}{y+5}$$

Rationalize the denominators.

①9 a) $\sqrt{\frac{20}{3}}$

b) $\sqrt{\frac{27}{40}}$

Subtract.

②0 $\frac{3}{x^2-5x+6} - \frac{2}{x^2+2x-8}$

Add and/or Subtract.

②1 $\frac{1}{x+6} - \frac{x+1}{x-6} + \frac{7x^2}{x^2-36}$

Add and/or Subtract.

$$\textcircled{22} \frac{1}{2x^2+7x+6} - \frac{2x}{x+2} + \frac{5x}{2x+3}$$

Simplify.

$$\textcircled{23} \frac{\frac{-3}{x^2-4x-32}}{\frac{5}{2x^2-16x}}$$

$$\textcircled{24} \quad \frac{\frac{3}{x-3} - \frac{4}{x+9}}{\frac{-2x+1}{x^2+6x-27}} + 2$$

Add and/or subtract.

$$\textcircled{25} \quad \frac{x-7}{81x^2-16} - 1 + \frac{1}{4-9x}$$

Test II (Version 2) Answers

① $\frac{x+2}{x-8}$

② $2x-3$

③ $\frac{5x^2(x+3)}{4(x-2)}$ or $\frac{5x^3+15x^2}{4x-8}$

④ $\frac{1}{(3x-8y)(x-5)}$

⑤ $\frac{x-1}{5x^2}$

⑥ $\frac{-4x+12}{(x+4)(x+4)}$ or $\frac{12-4x}{(x+4)^2}$ or $\frac{12-4x}{x^2+8x+16}$

⑦ $\frac{60x-59}{14x(x-6)}$ or $\frac{60x-59}{14x^2-84x}$

⑧ a) $\frac{5}{7}$ b) 8 c) 2.646 d) 3

⑨ a) $2\sqrt{15}$ b) $10\sqrt{2}$ c) $-3\sqrt{2}$ d) $-6\sqrt{11}$

⑩ a) 5 b) $-5\sqrt{42}$ c) $-8\sqrt{30}$ d) $-15\sqrt{6}+24-6\sqrt{5}$

⑪ a) $6\sqrt{35}$ b) 5 c) $\frac{3\sqrt{2}}{2}$ d) $\frac{3\sqrt{10}}{10}$

⑫ a) $\frac{27}{32}$ b) $\frac{7}{40}$ c) $-\frac{80}{3x}$ d) 15

⑬ $-\frac{(2x^2-1)(2x^2-3)}{8x^{10}}$ or $-\frac{4x^4-8x^2+3}{8x^{10}}$

⑭ a) $4\sqrt{2}-8\sqrt{5}-20\sqrt{3}$ b) $2\sqrt{35}$

⑮ a) $\frac{1}{4}$ b) $\frac{\sqrt{42}}{7}$ c) $\frac{2\sqrt{6}}{3}$ d) $\frac{\sqrt{11}}{4}$

⑯ a) $\frac{6}{x(x-3)}$ or $\frac{6}{x^2-3x}$ b) $\frac{\sqrt{42}}{21}$

⑰ $\frac{16x^2y}{15z}$

⑱ $\frac{-22y-135}{5y(y+5)}$ or $\frac{-22y-135}{5y^2+25y}$ or $-\frac{22y+135}{5y^2+25y}$

(19) a) $\frac{2\sqrt{15}}{3}$ b) $\frac{3\sqrt{30}}{20}$

(20) $\frac{x+18}{(x-3)(x-2)(x+4)}$

(21) $\frac{6x^2-6x-12}{(x+6)(x-6)}$ or $\frac{6x^2-6x-12}{x^2-36}$

(22) $\frac{x^2+4x+1}{(2x+3)(x+2)}$ or $\frac{x^2+4x+1}{2x^2+7x+6}$

(23) $\frac{-6x}{5(x+4)}$ or $\frac{-6x}{5x+20}$ or $-\frac{6x}{5x+20}$

(24) $\frac{-x+39}{2x^2+10x-53}$

(25) $\frac{-81x^2-8x+5}{(9x+4)(9x-4)}$ or $\frac{-81x^2-8x+5}{81x^2-16}$